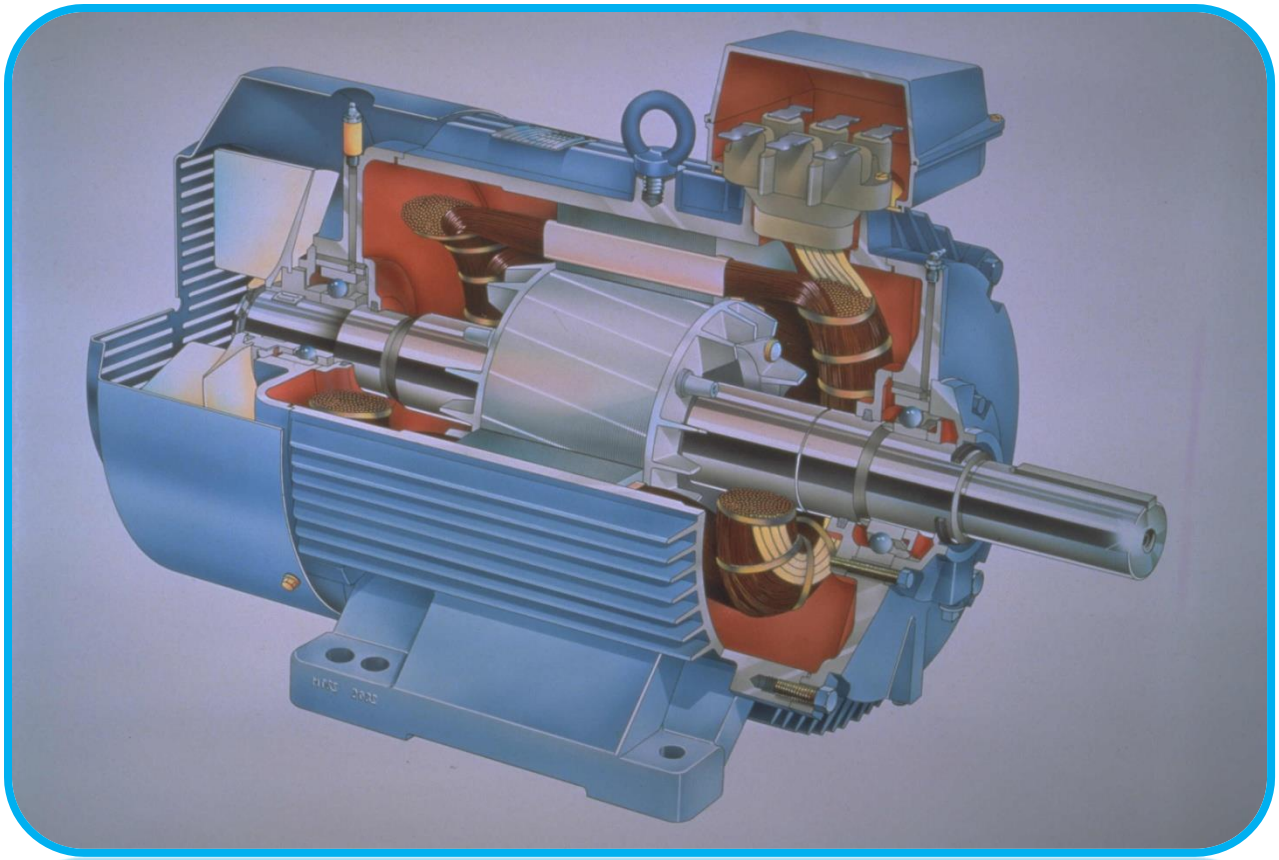
Module 4

Three phase induction Motor

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Three phase induction Motor



Module IV

4	3-phase induction motor – principle of operation - classification - constructional features – torque equation - torque-slip characteristics – numerical problems (3 hours) Phasor diagram - equivalent circuit (1 hour) Power flow diagram - losses and efficiency – numerical problems (1 hour) No-load and blocked-rotor tests – determination of equivalent circuit parameters – numerical problems (1 hour) Starting of induction motors – types of starters – DOL starter, autotransformer starter, star-delta starter, rotor resistance starter (no design) – numerical problems (3 hours) Single-phase induction motors – double revolving field theory –torque slip characteristics – types – split-phase, capacitor-start induction-run, permanentcapacitor types – applications (2 hours)	11



Introduction:

- Generation of AC supply is easier.
- Working based on electro magnetic inducti on.
- It is called as rotating transformer.
- Common and frequently used in Industries.



Advantages :

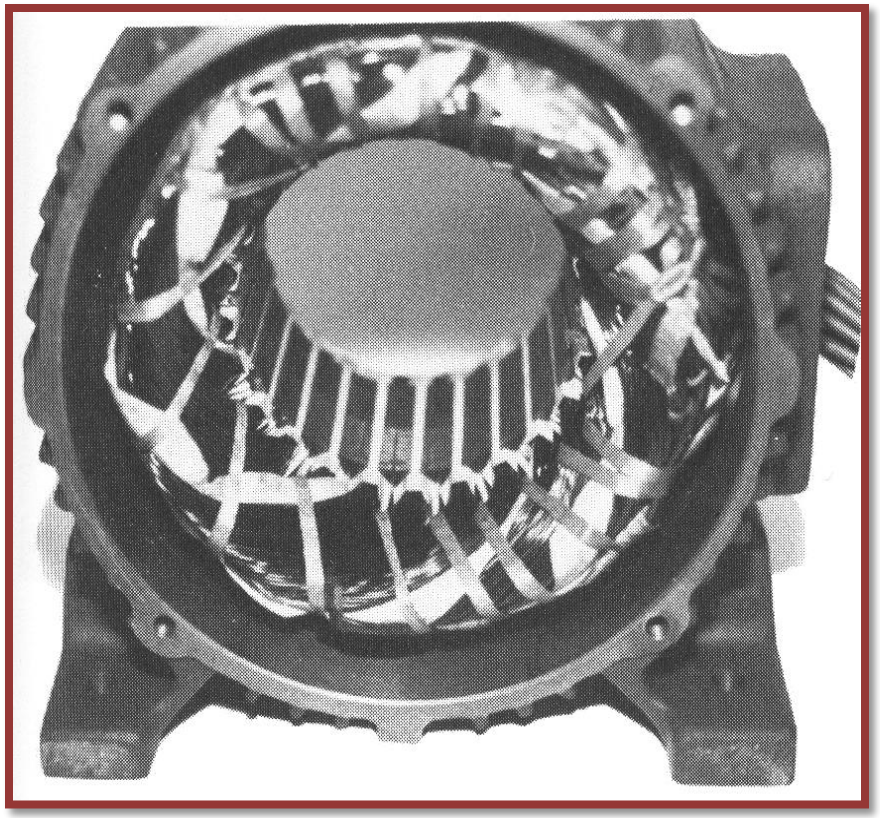
1. Simple construction
2. High reliability
3. Low cost
4. High efficiency
5. Less maintenance
6. Self starting
7. Good power factor

Constructional Details

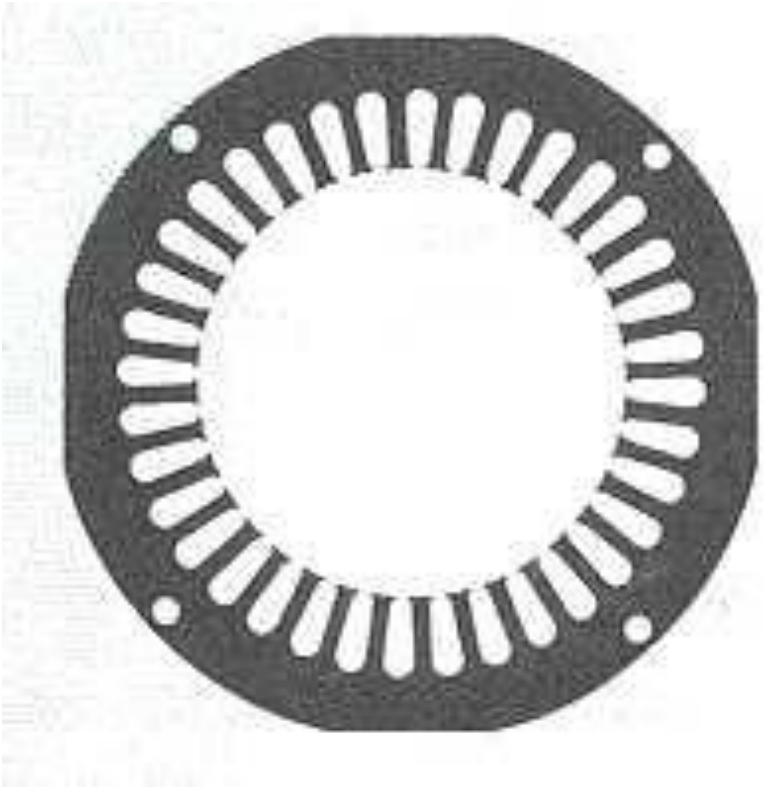
Two main parts of induction Motor - **Stator & Rotor**

stator

- consisting of a st eel frame that sup ports a hollow, cy lindrical core
- core, constructed from stacked lami nations (why?), h aving a number of evenly spaced slot s, providing the s pace for the stator winding



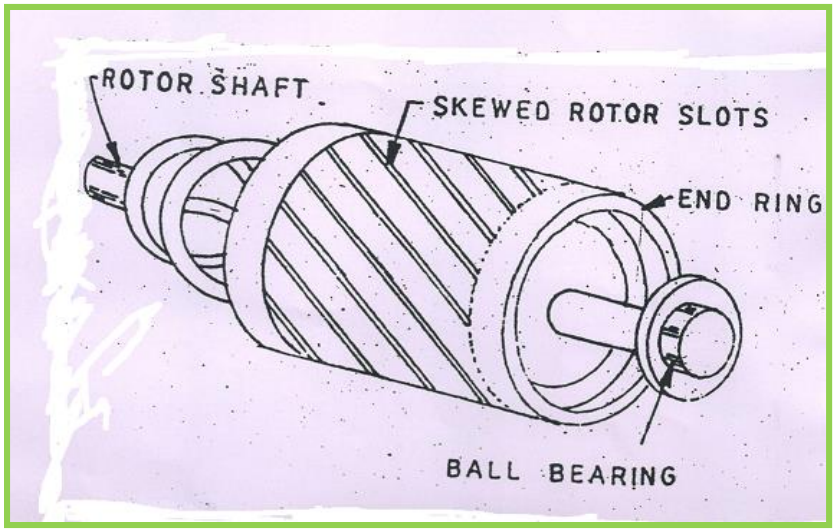
- **The elements of the laminated iron core are punched from a silicon iron sheet.**



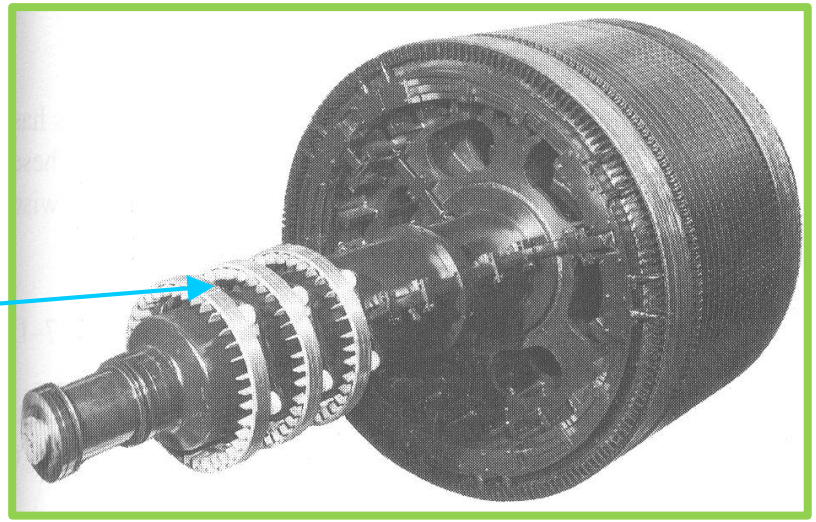
Stator iron core sheet.

Rotor

Squirrel cage rotor

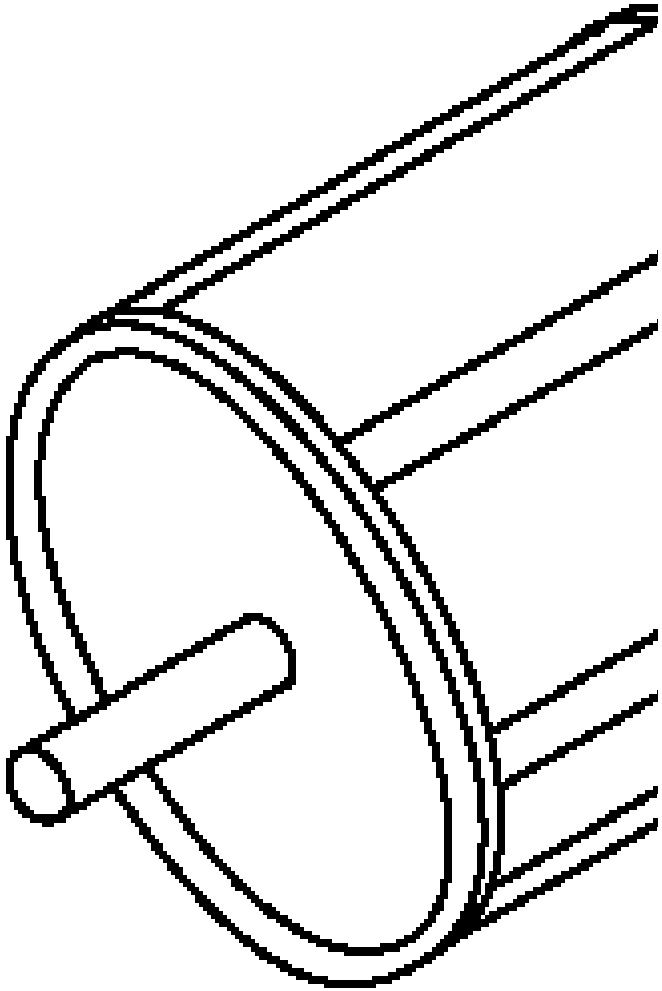


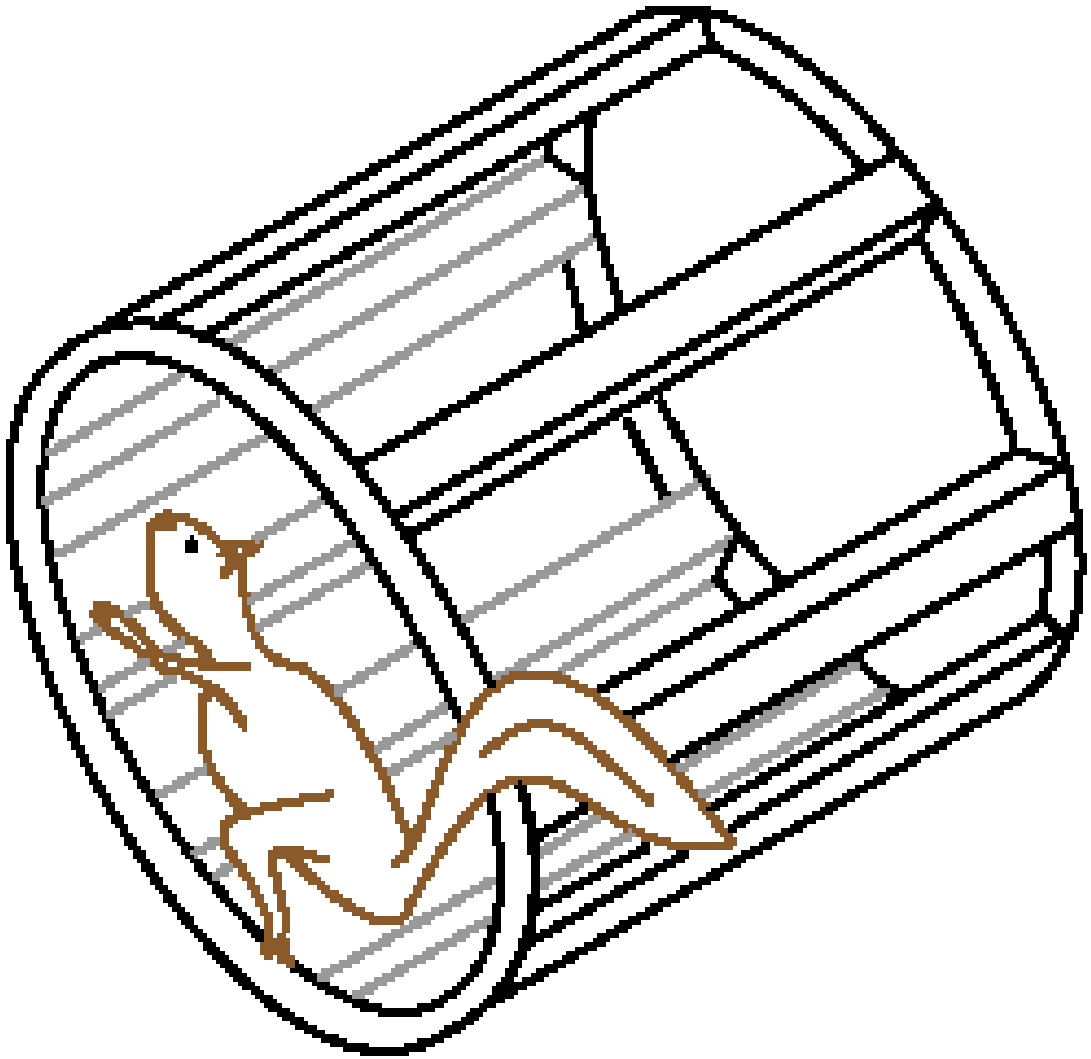
Wound rotor



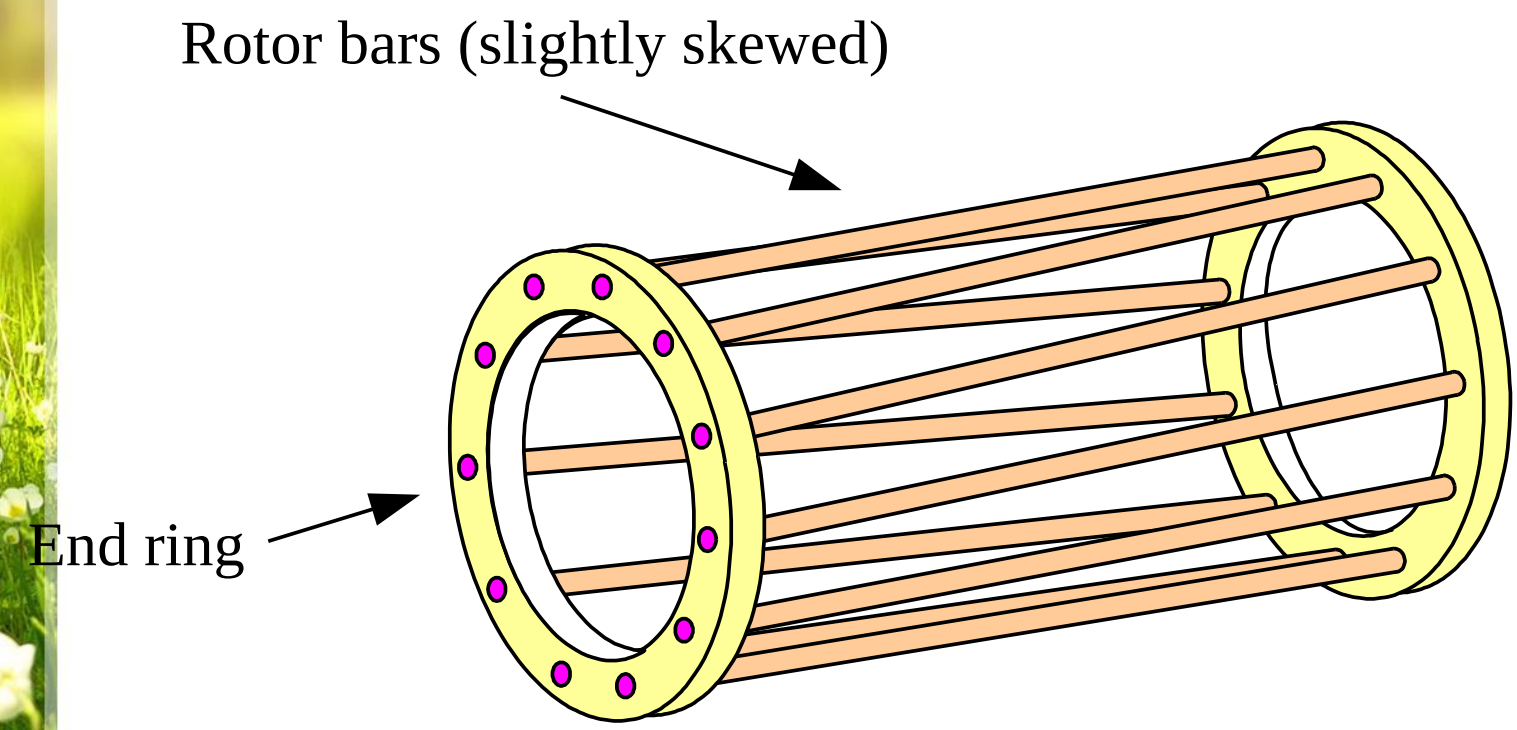
Notice the slip rings

Squirrel Cage Induction motor





Induction Motors



Induction Motors

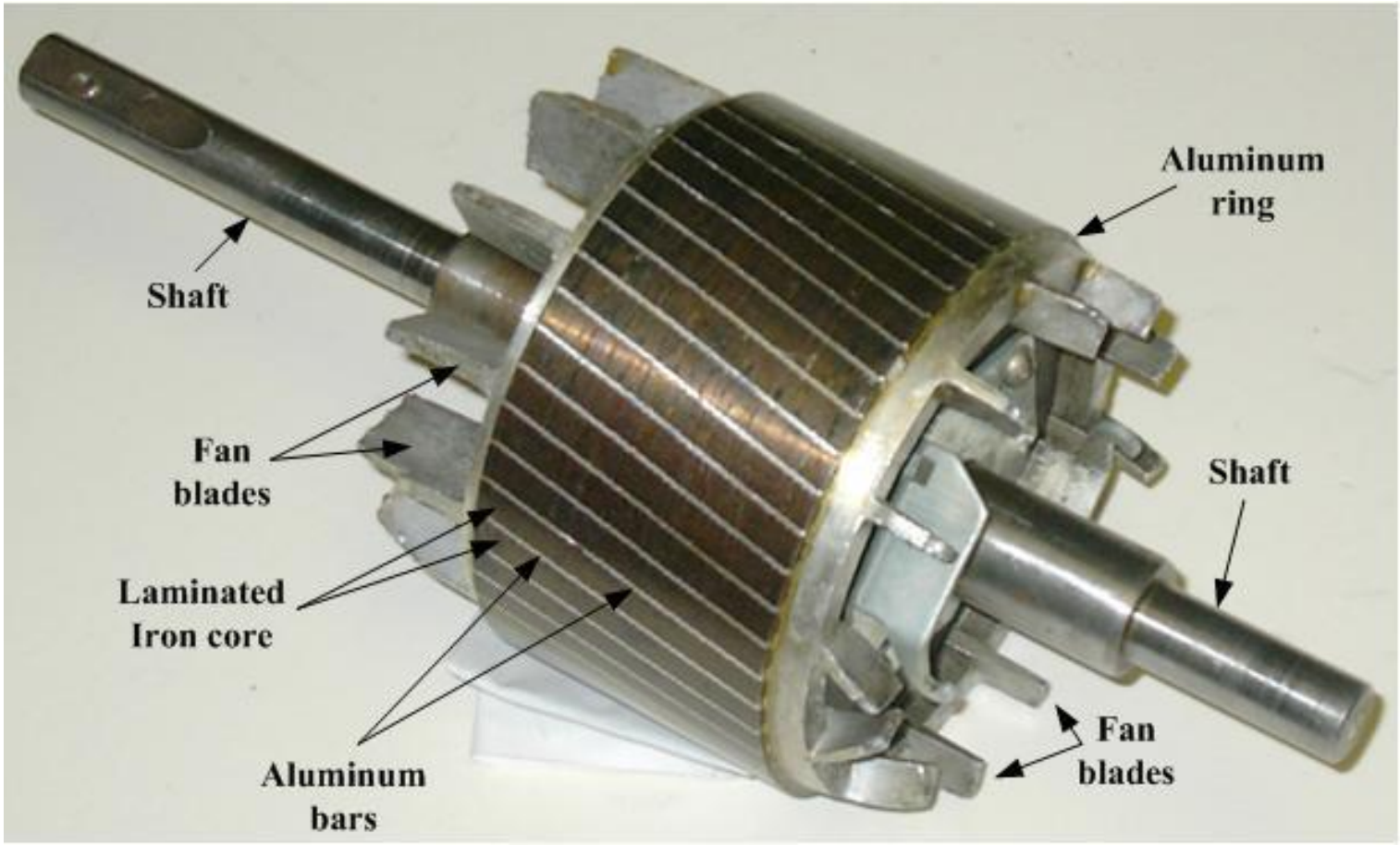
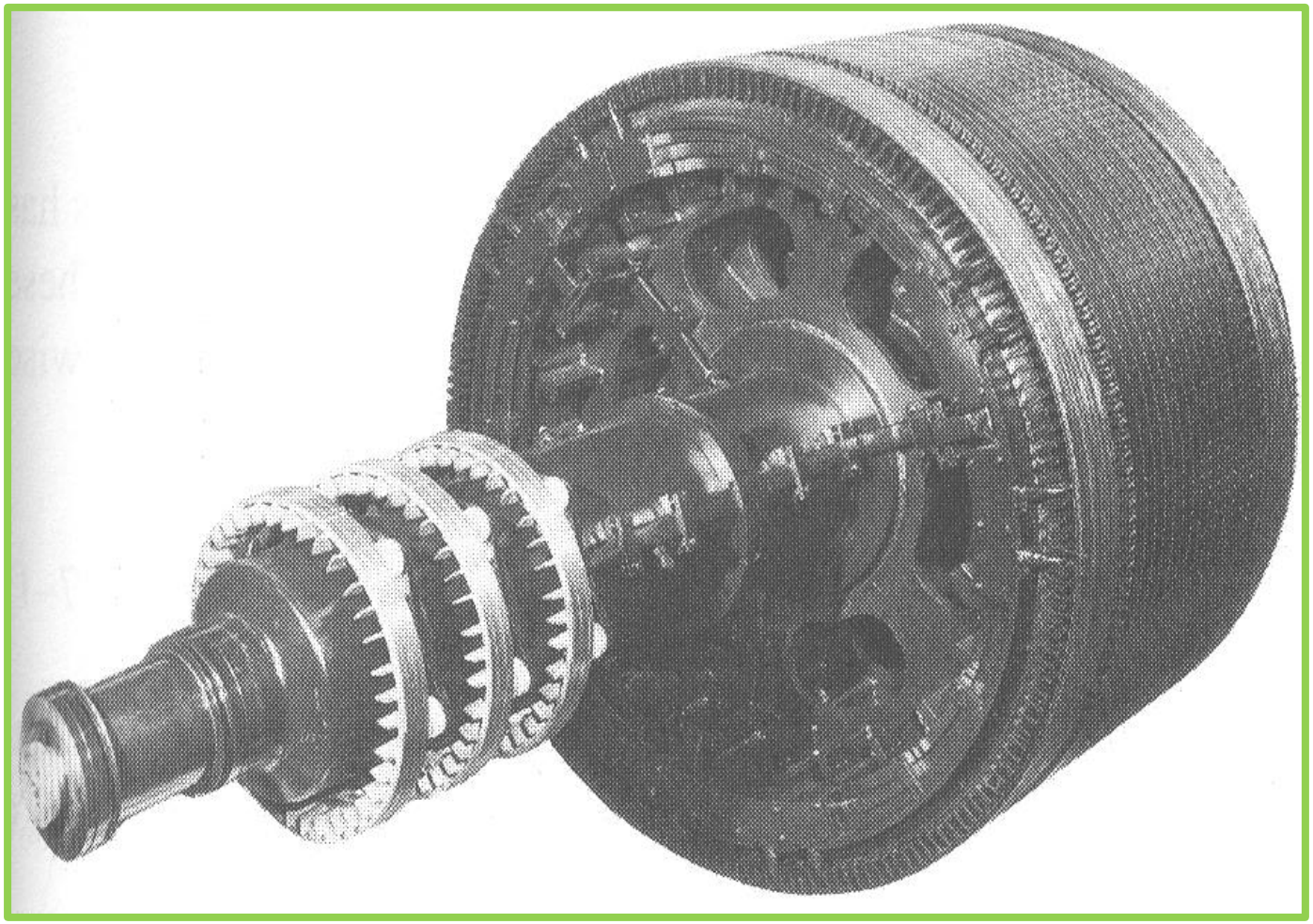


Figure 7.10 Squirrel cage rotor.

Wound rotor IM

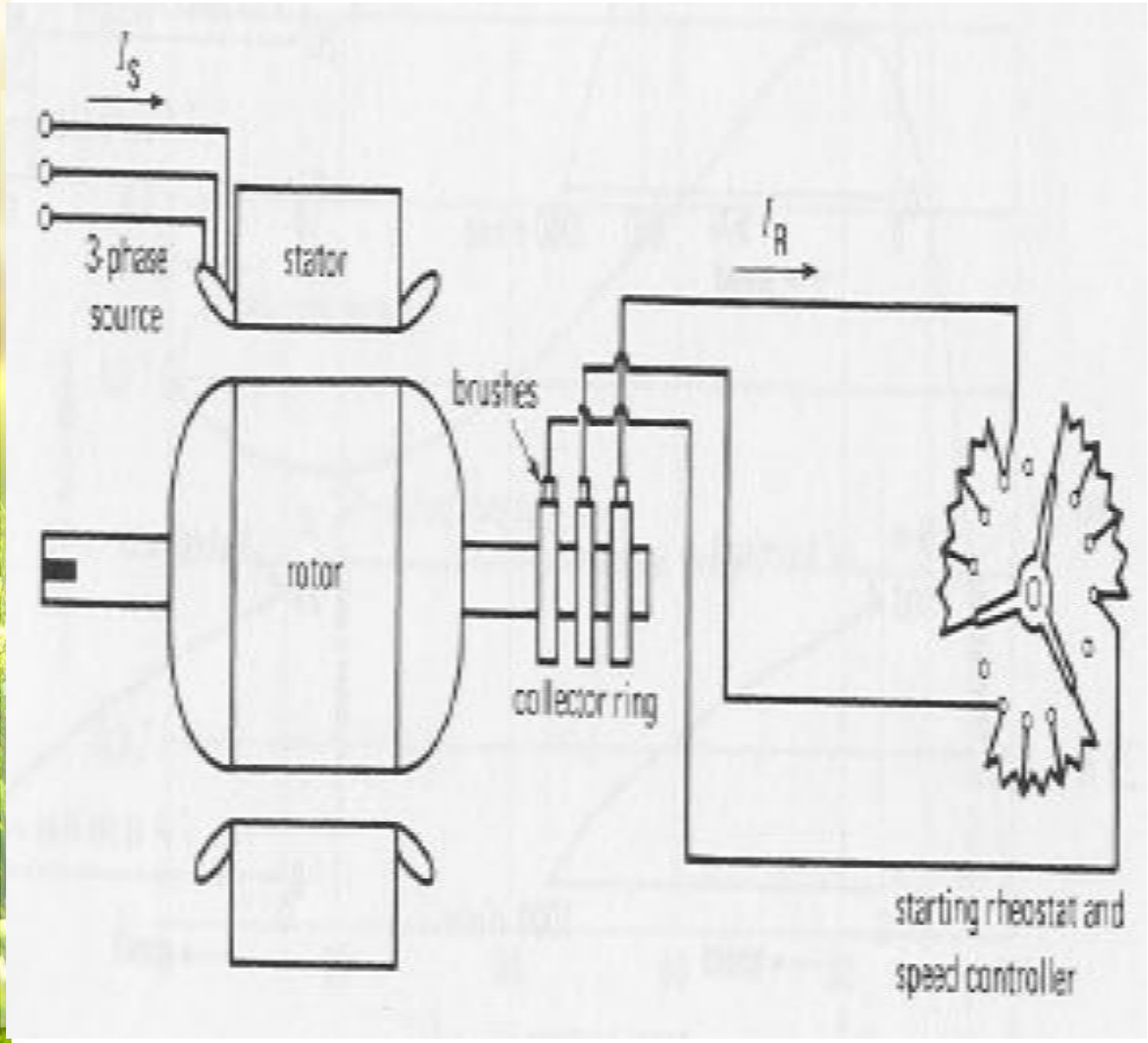




- A wound rotor has a 3-phase winding, similar to the stator winding.

- The rotor winding terminals are connected to three slip rings which turn with the rotor. The slip rings/brushes allow external resistors to be connected in series with the winding.

- The external resistors are mainly used during start-up under normal running conditions the windings short circuited externally.


I_S
3-phase source
stator
I_R
brushes
rotor
collector ring
starting rheostat and speed controller



Two basic design types of rotor:

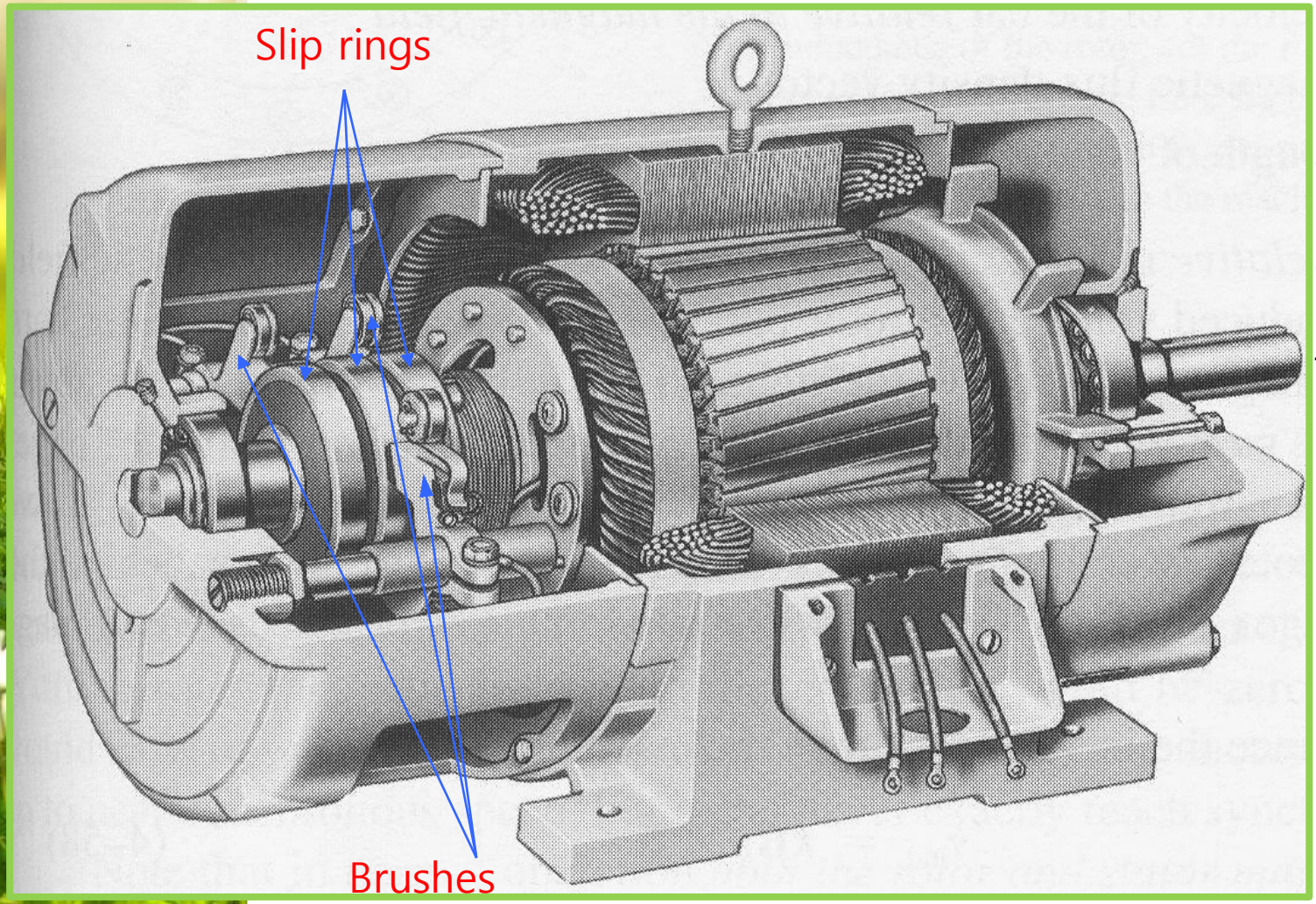
Squirrel-cage:

Conducting bars laid into slots and shorted at both ends by shorting rings.

Wound-rotor:

- Complete set of 3Φ windings usually Y-connected, the ends of the rotor wires are connected to 3 slip rings on th e rotor shaft.
- External resistance is connected in each winding at start ing. So it reduce the starting current and improve the p.f of the rotor
- Higher rotor resistance at starting increases the starting torque.
- As the motor speeds up resistance in the rotor circuit is cut step by step and finally the rotor is short circuited .

Construction

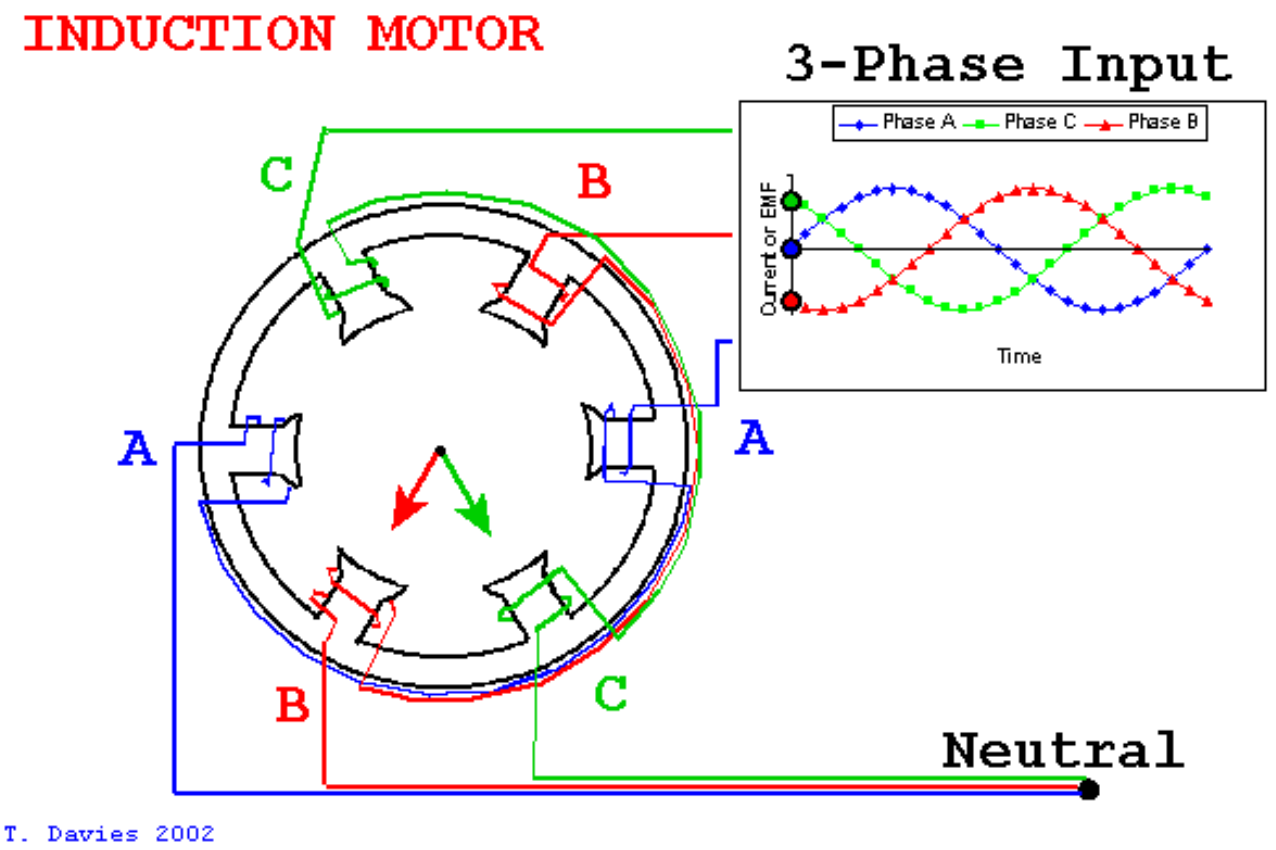


Cutaway in a typical wound-rotor IM. Notice the brushes and the slip rings

Types of Induction motor

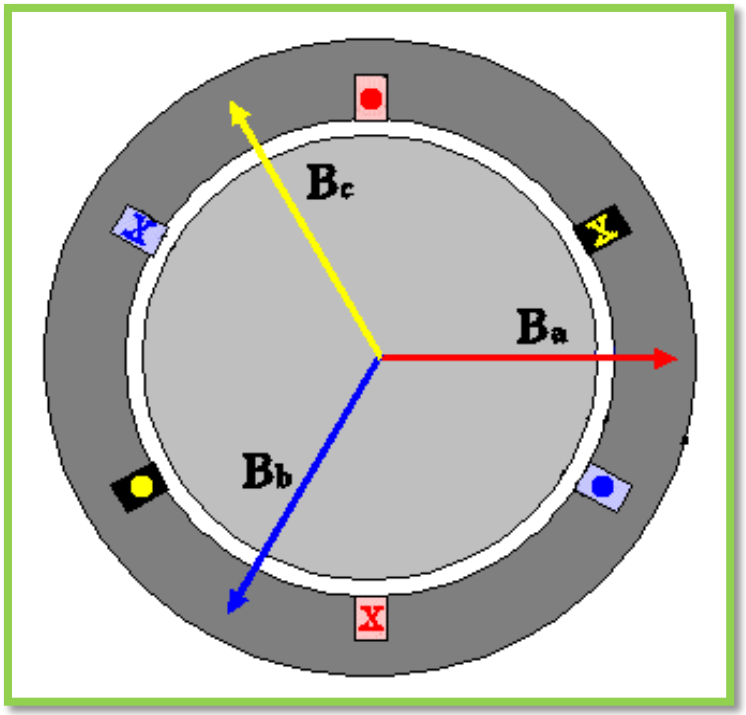
Sl.No	Squirrel cage motor	Slip ring motor
1	Construction is very simple	Rotor has winding . So the construction is not simple
2	Low cost	High cost
3	Less starting torque	High starting torque
4	High efficiency	Low efficiency
5	power factor is less	Power factor is high
6	High starting current	Low starting current
7	No slip ring and no sparks	Brushes on the slip ring makes spark

Principle of operation



Rotating Magnetic Field

- Balanced three phase wi ndings, i.e. mechanically displaced 120 degrees fo rm each other, fed by bal anced three phase source





- When all the three phases are en ergised simultaneously, the sum o f the three fields produced by the windings will have a constant ma gnitude and rotating with consta nt angular velocity.

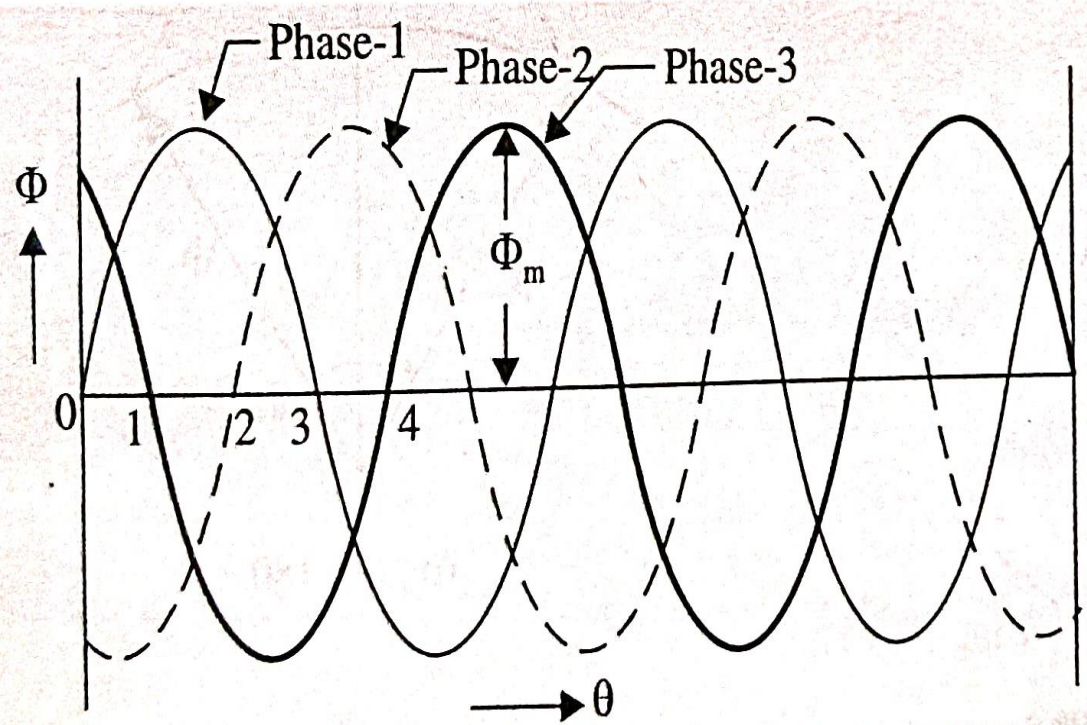


Fig. 34.12

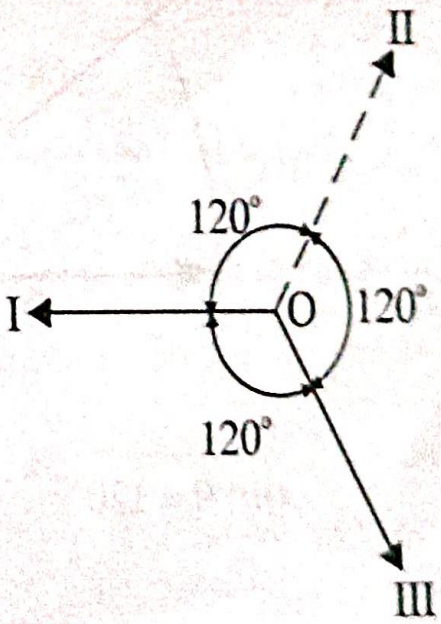
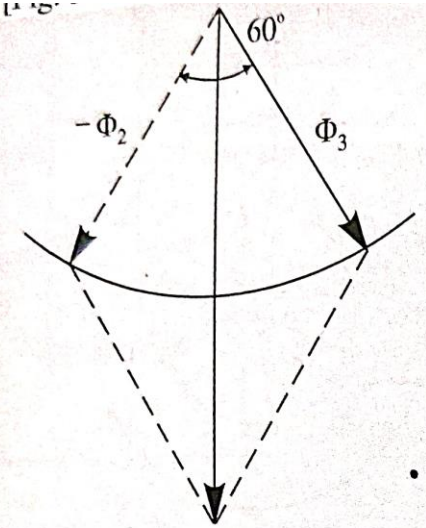
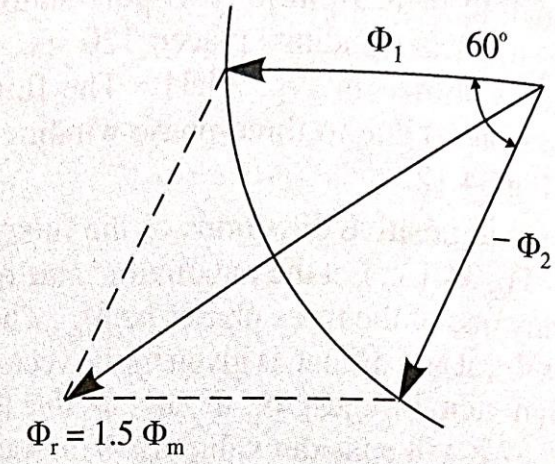


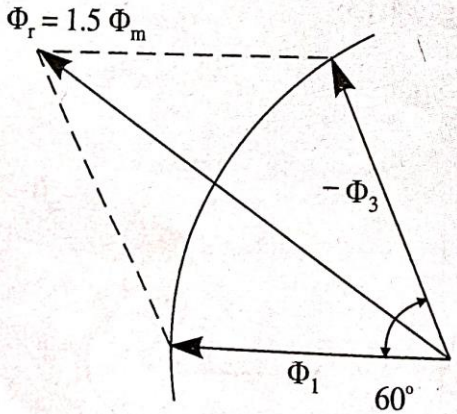
Fig. 34.13



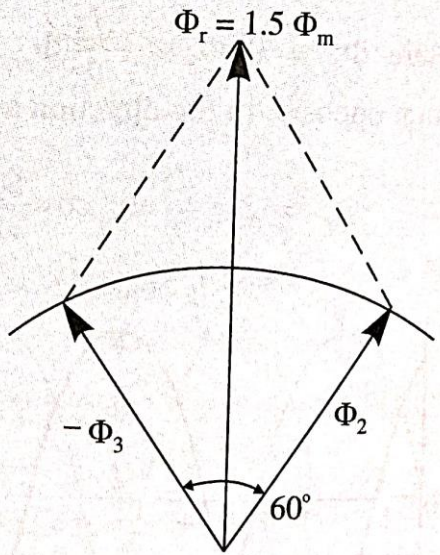
(*i*) $\theta = 0°$



(*ii*) $\theta = 60°$



(*iii*) $\theta = 120°$



(*iv*) $\theta = 180°$

(*i*) When $\theta = 0°$ *i.e.* corresponding to point *O* in Fig. 34.12

Here $\Phi_1 = 0, \Phi_2 = -\frac{\sqrt{3}}{2}\Phi_m, \Phi_3 = \frac{\sqrt{3}}{2}\Phi_m$. The vector for Φ_2 in Fig. 34.14 (*i*) is drawn in a direction opposite to the direction assumed positive in Fig. 34.13.

$$\therefore \quad \Phi_r = 2 \times \frac{\sqrt{3}}{2}\Phi_m \cos\frac{60°}{2} = \sqrt{3} \times \frac{\sqrt{3}}{2}\Phi_m = \frac{3}{2}\Phi_m$$

(*ii*) when $\theta = 60°$ *i.e.* corresponding to point 1 in Fig. 34.12.

Here $\Phi_1 = \frac{\sqrt{3}}{2}\Phi_m$ ...drawn parallel to *OI* of Fig. 34.13 as shown in Fig. 34.14 (*ii*)

$\Phi_2 = -\frac{\sqrt{3}}{2}\Phi_m$ ...drawn in opposition to *OII* of Fig. 34.13.

$\Phi_3 = 0$

$$\therefore \quad \Phi_r = 2 \times \frac{\sqrt{3}}{2}\Phi_m \times \cos 30° = \frac{3}{2}\Phi_m$$ [Fig. 34.14 (*ii*)]



Three phase supply

- Produces rotating magnetic field
- Resultant flux is constant and 1.5 times of max value of flux in one p hase
- Flux rotates around the stator at synchronous speed

$$n_{sync} = \frac{120 f_e}{P} \quad rpm$$

- A rotating magnetic field with constant magn itude is produced, rotating with a speed

$$n_{sync} = \frac{120\, f_e}{P} \quad rpm$$

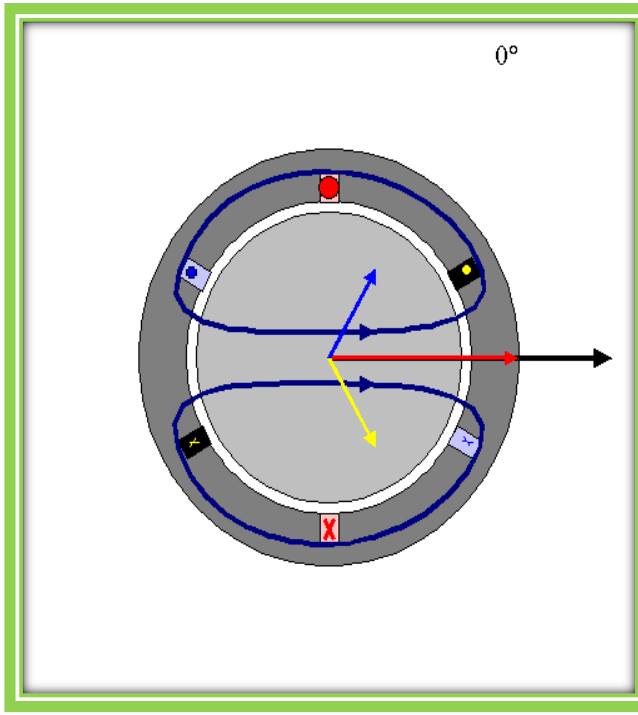
Where f_e is the supply frequency and
P is the no. of poles and
n_{sync} is called the synchronous speed in rpm

No of Poles and Synchronous Speed

P	50 Hz
2	3000
4	1500
6	1000
8	750
10	600
12	500

Rotating Field – Principle of operation

- The phase waveforms in the sequence A-B-C. This produces a clockwise rotating magnetic field.
- If we interchange any two of the lines connected to the stator, the new phase sequence will be A-C-B.This will produce a counterclockwise rotating field, reversing the motor direction.





- This stationary conductor cut the rotating magnetic field of the stator, and because of the electromagnetic induction, the EMF induces in the rotor.
- This EMF is known as the rotor induced EMF.
- The conductors of the rotor are short-circuited either by the end rings or by the help of the external resistance.

- The relative motion between the rot ating magnetic field and the rotor c onductor induces the current in the rotor conductors. As the current flow s through the conductor opposes its cause (Lenz's law)
- Hence The rotor starts to rotate at a speed *N, which is slightly less than*
- the synchronous speed, *Ns*

Induction Motors Slip

The difference between the synchronous speed and rotor speed can be expressed as a percent age of synchronous speed, known as the slip:

where $$s = \frac{Ns - N}{Ns} \times 100\ \%$$

s = slip,

Ns = synchronous speed (rpm),

N = rotor speed (rpm)

Frequency Induced In the Rotor

The frequency induced in the rotor depends on the slip:

- When the rotor is blocked (s=1) , the frequency of the in duced voltage is equal to the supply frequency
- If the rotor runs at synchronous speed (s = 0), the freque ncy will be zero

$$f_R = sf$$

where

f_R = frequency of voltage and current in the rotor,

f = frequency of the supply

s = slip

Induction motor speed

- So, the IM will always run at a speed low er than the synchronous speed
- The difference between the motor speed a nd the synchronous speed is called the Sli p

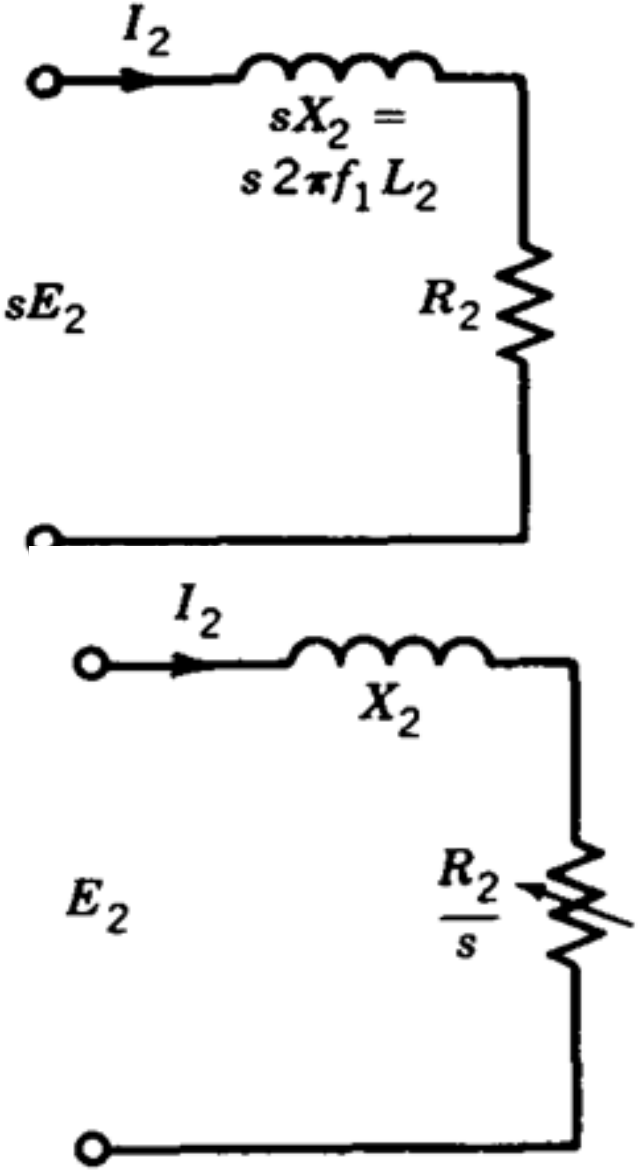
$$n_{slip} = n_{sync} - n_m$$

Where n_{slip}= slip speed

n_{sync}= speed of the magnetic field

n_m = mechanical shaft speed of the mot or

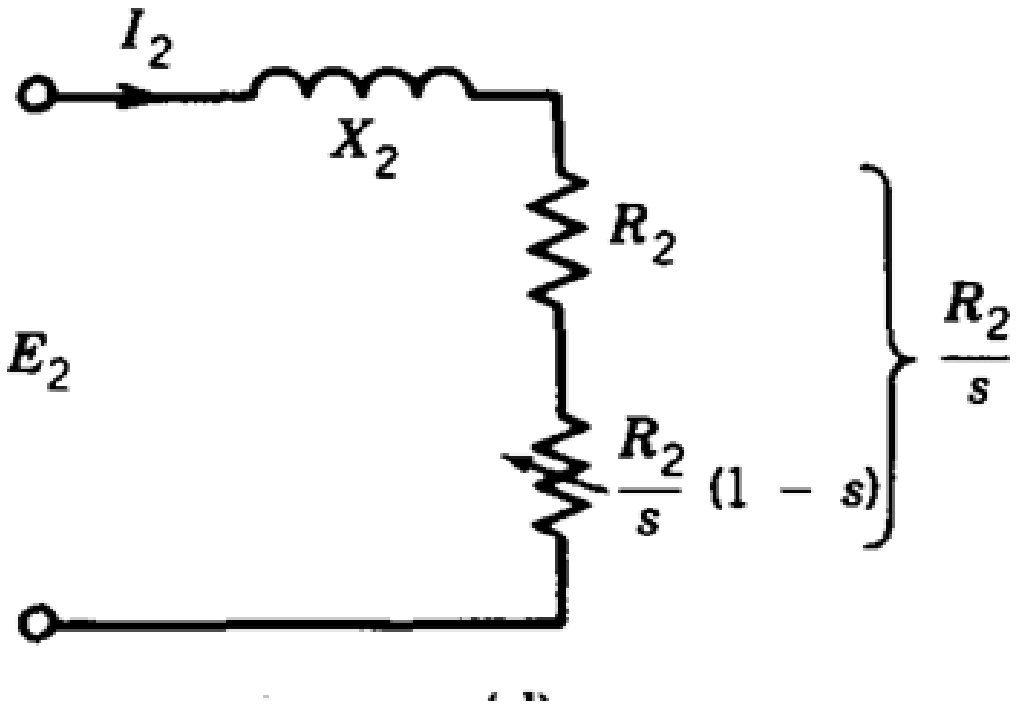
Rotor Current



$$I_2 = \frac{sE_2}{R_2 + jsX_2}$$

$$I_2 = \frac{E_2}{(R_2/s) + jX_2}$$

Rotor equivalent circuit



Rotor Torque

Torque depends on

- Flux per pole
- Rotor current
- Phase angle between flux and rotor current

$$T \propto \emptyset\, I_2\, Cos\, \emptyset_2$$

$$E_2 \propto \emptyset$$

$$T \propto E_2\ I_2\, Cos\, \emptyset_2$$



Where $$I_2 = \frac{E_2}{(R_2/s) + jX_2}$$

Substituting in the above torque eq uation,

$$T \propto E_2 \frac{sE_2}{\sqrt{R_2^2 + sX_2^2}} \frac{R_2}{\sqrt{R_2^2 + sX_2^2}}$$

$$T \propto \frac{s\,E_2^{\,2} R_2}{(R_2^2 + sX_2^2)}$$

Condition for maximum torque

- To get maximum value differentiate torque equation with R2

$$T \propto \frac{s\,{E_2}^2 R_2}{({R_2}^2 + (sX_2)^2)\cdot}$$

$$\frac{dT}{dR_2} = 0$$

- Then we will get, $s = \frac{R_2}{X_2}$
- Maximum torque $T_{max} = \frac{K{E_2}^2}{2X_2}$

- Running torque (slip =s)

$$T = k\frac{s\,{E_2}^2 R_2}{({R_2}^2 + (sX_2)^2)\cdot}$$

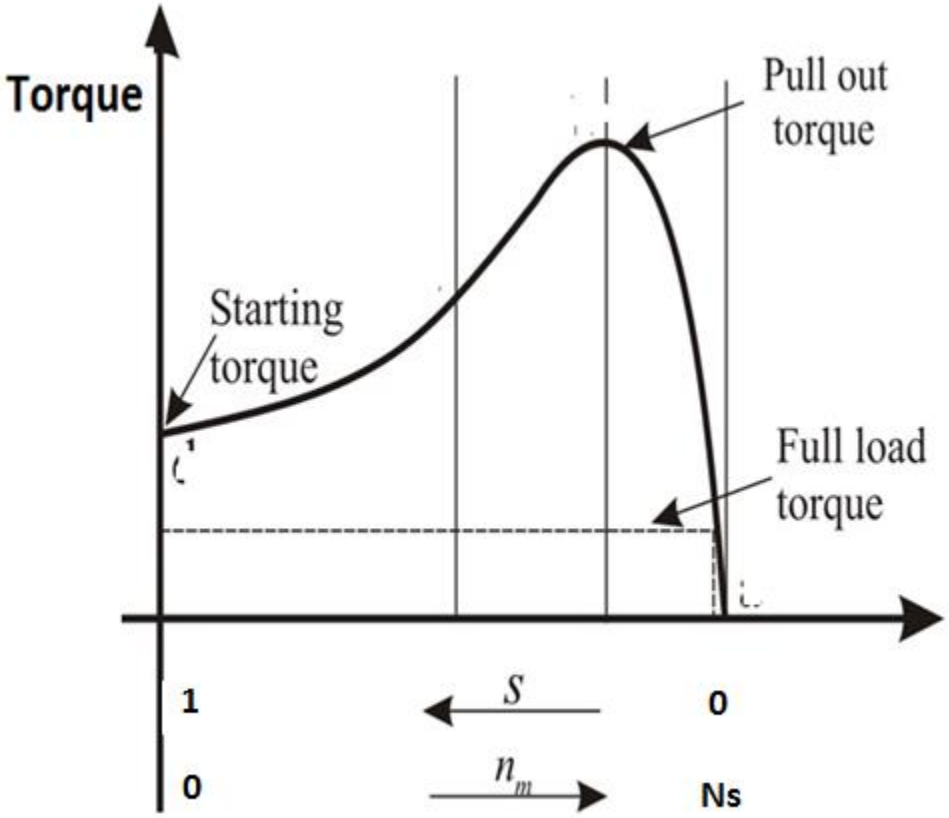
- Starting torque (slip = 1)

$$T_{st} = k\frac{{E_2}^2 R_2}{({R_2}^2 + {X_2}^2)\cdot}$$

- Maximum torque

$$T_{max} = \frac{K{E_2}^2}{2X_2}$$

Torque-speed characteristics
or
Torque-slip characteristics



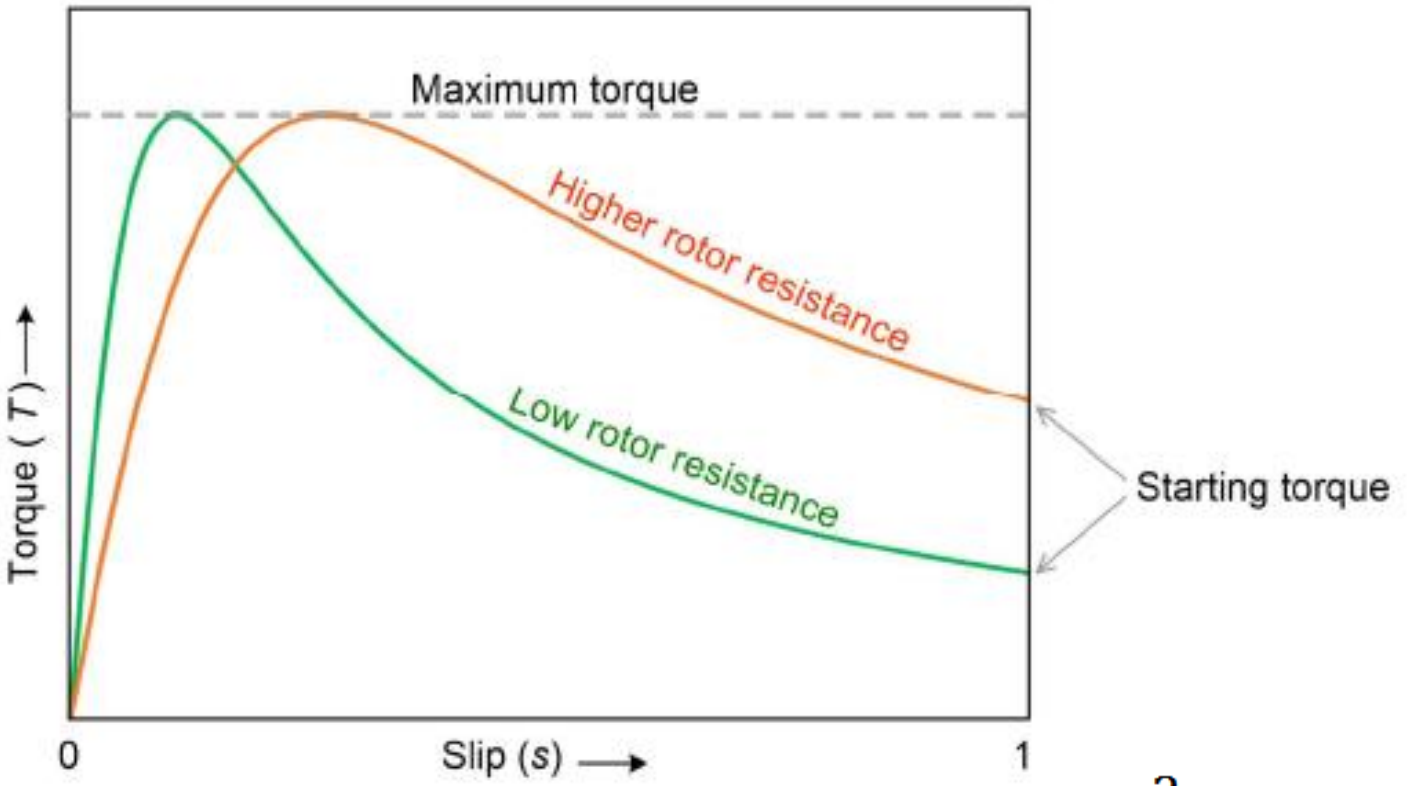
- For lower values of slip $(sx_2)^2$ is very low ie,

$$T \propto s$$

- For higher values of slip $(sx_2)^2$ is very large ie,

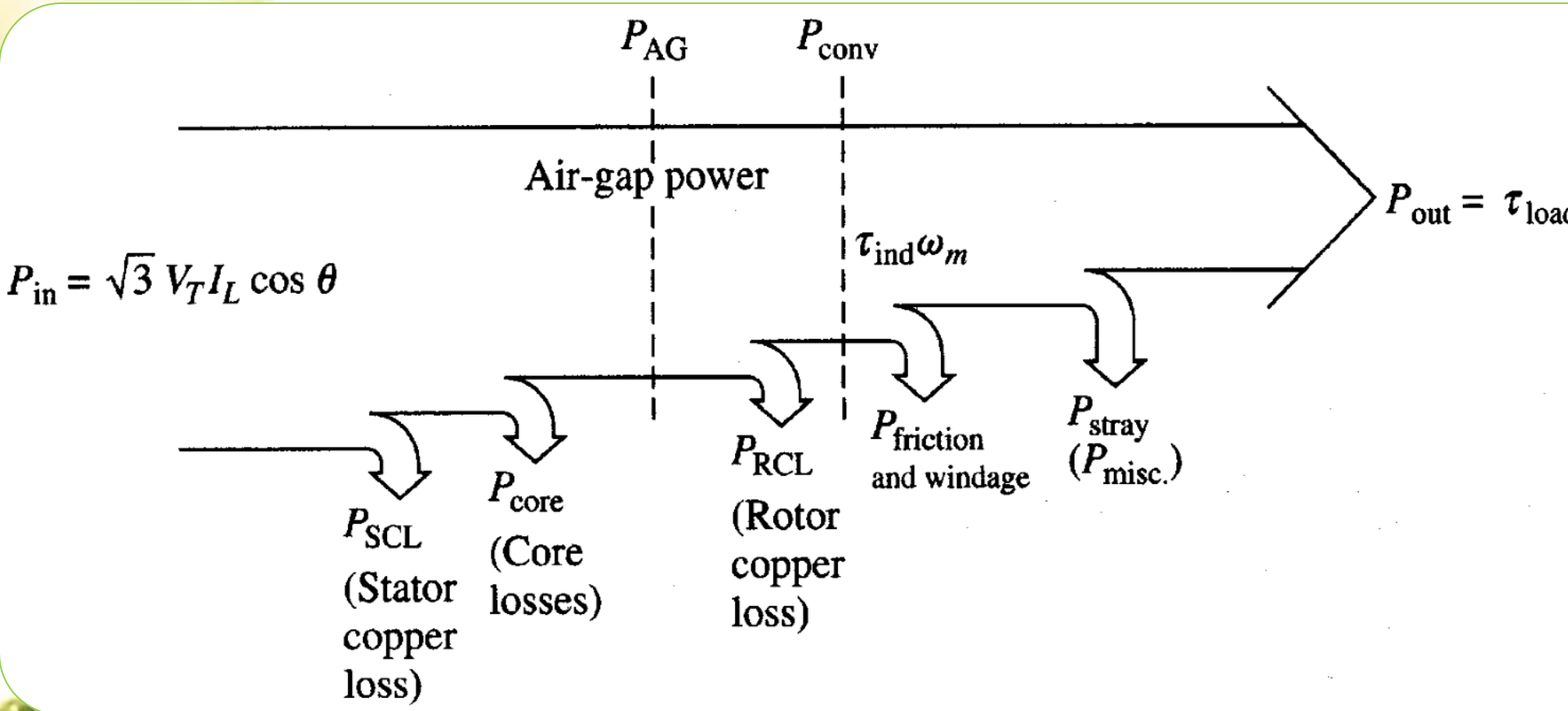
$$T \propto \frac{1}{s}$$

Variation of torque slip characteristics with change in rotor resistance



$$T \propto \frac{s\,E_2{}^2 R_2}{(R_2{}^2 + sX_2{}^2)}$$

Power flow in induction motor





Copper losses

Copper loss in the stator (P_{SCL}) = $I_1^2 R_1$

Copper loss in the rotor (P_{RCL}) = $I_2^2 R_2$

Power relations

$$P_{in} = \sqrt{3}\, V_L I_L \cos\theta = 3 V_{ph} I_{ph} \cos\theta$$

$$P_{SCL} = 3 I_1^2 R_1$$

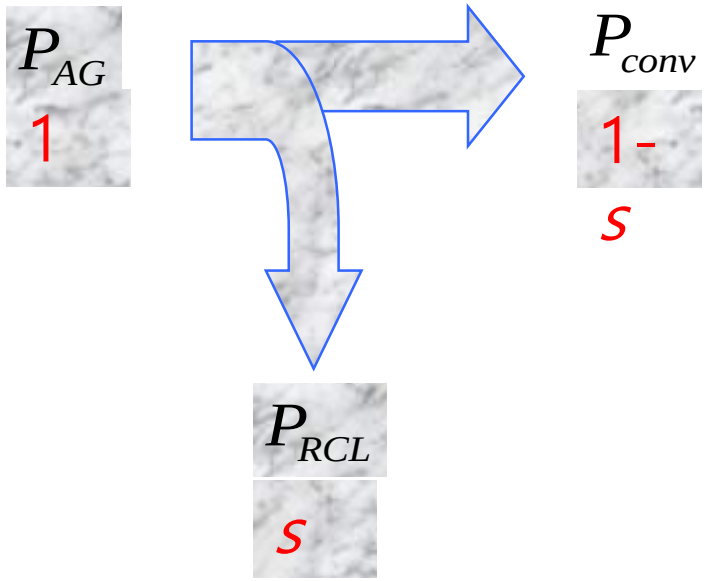
$$P_{AG} = P_{in} - (P_{SCL} + P_{core})$$

$$P_{RCL} = 3 I_2^2 R_2$$

$$P_{conv} = P_{AG} - P_{RCL}$$

$$P_{out} = P_{conv} - (P_{f+w} + P_{stray})$$

$$\tau_{ind} = \frac{P_{conv}}{\omega_m}$$



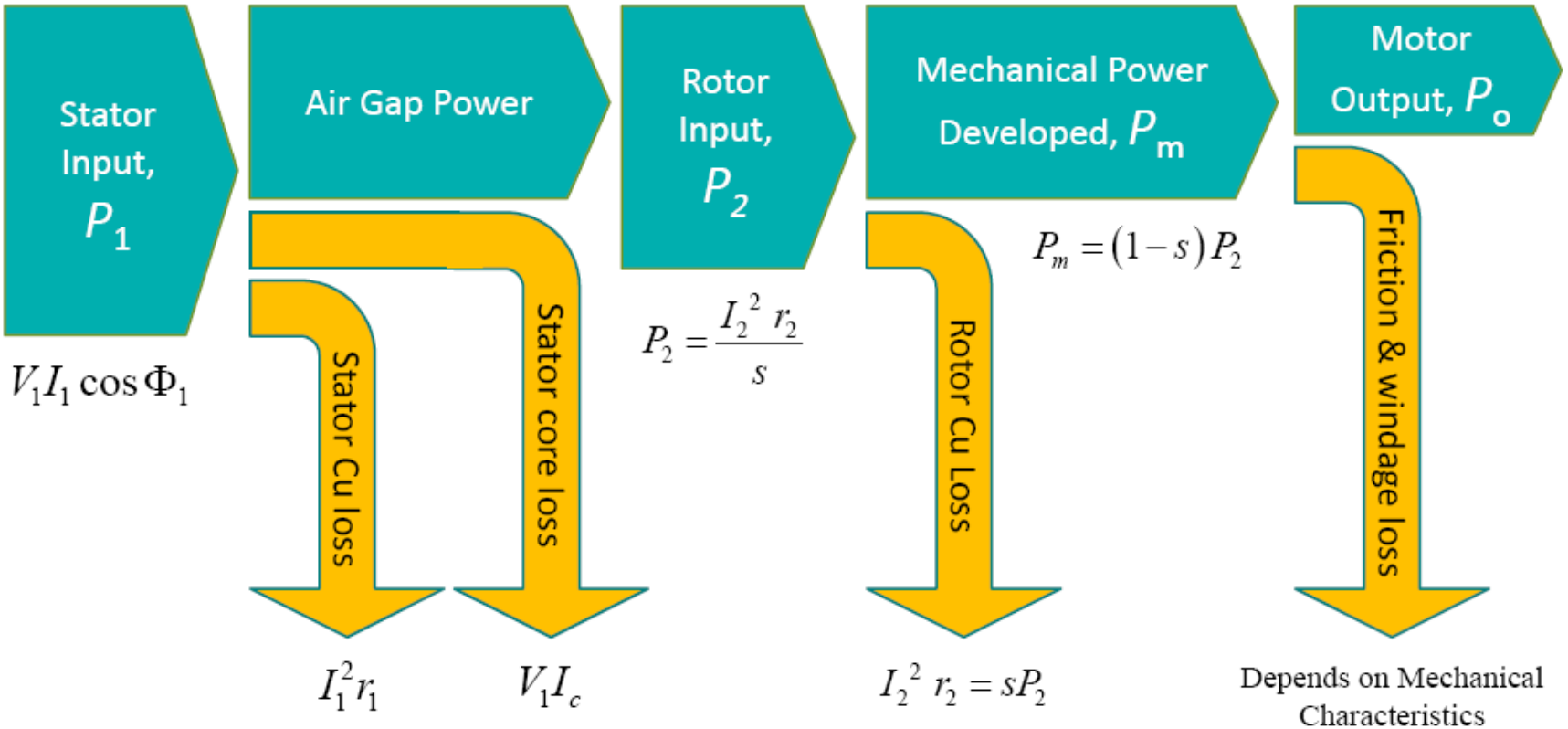
$$P_{AG} : P_{RCL} : P_{conv}$$

$$1 : s : 1\text{-}s$$

Power Flow

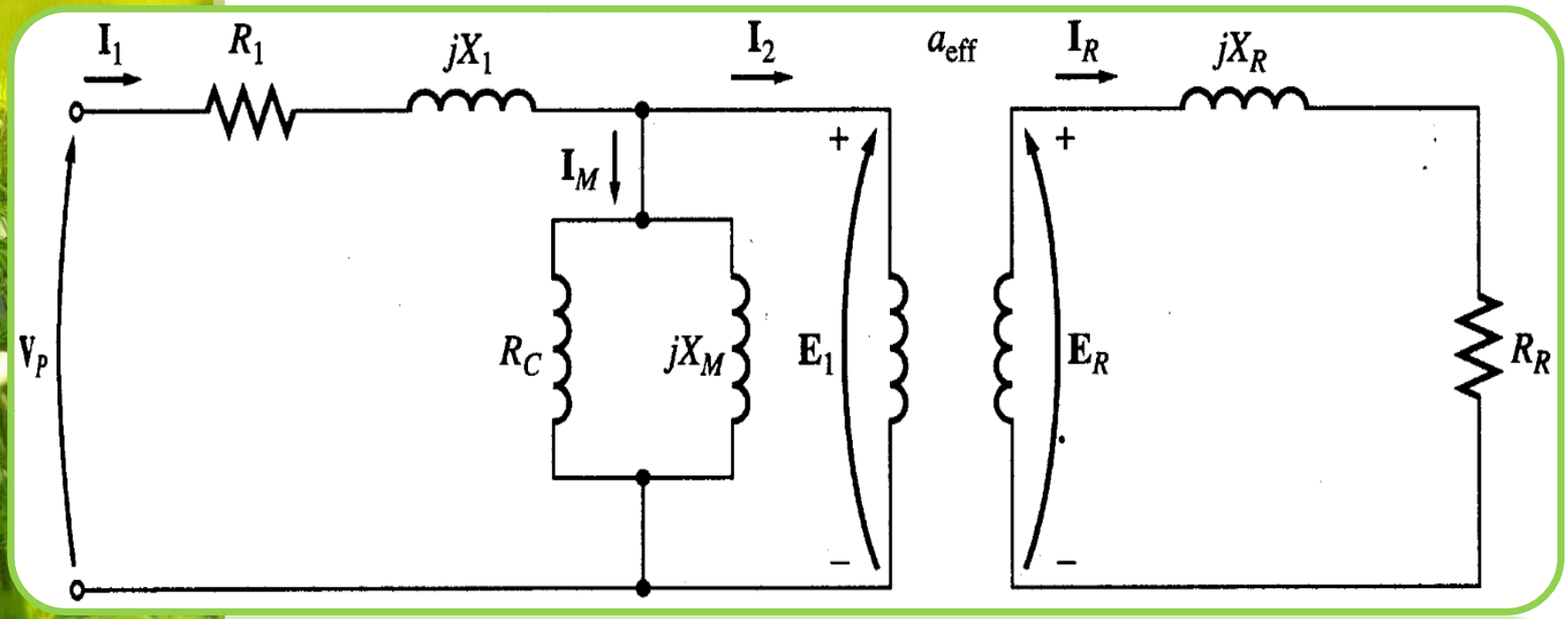
Power Flow Diagram

All Parameters are per phase values



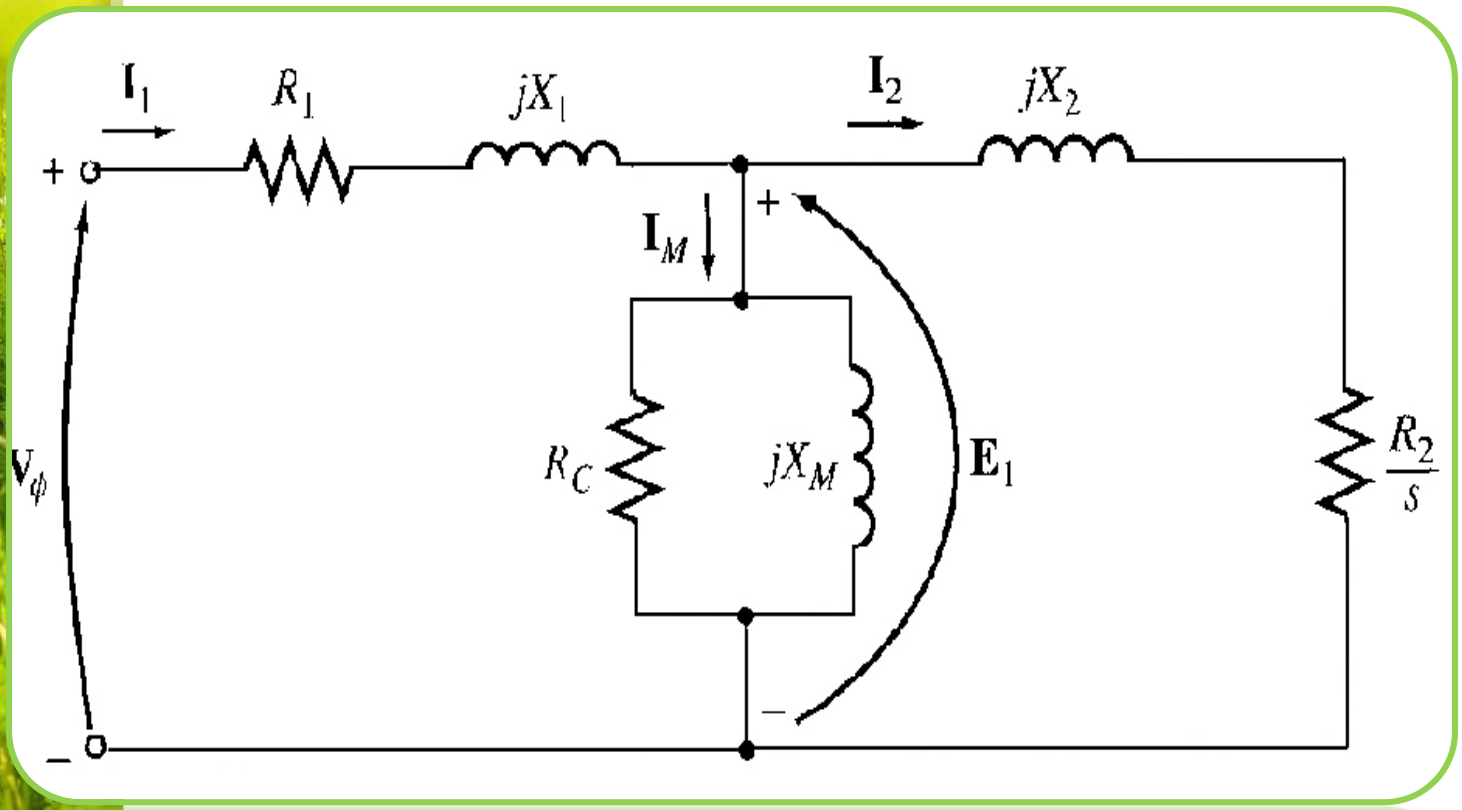
Equivalent Circuit

The induction motor is similar to the transformer with the ex ception that its secondary windings are free to rotate

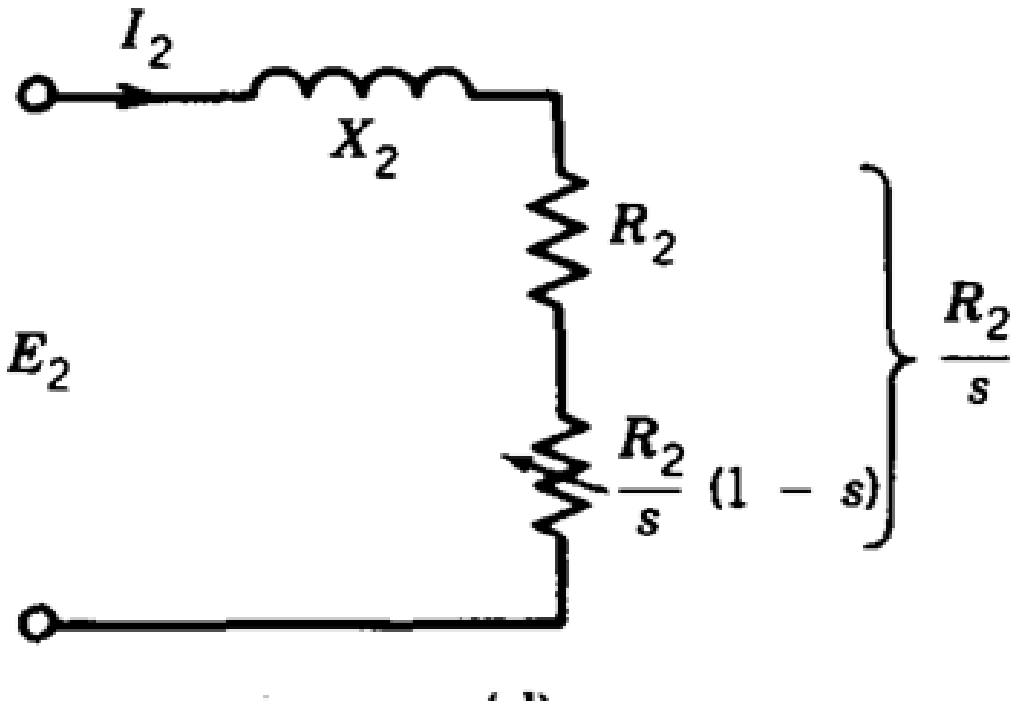


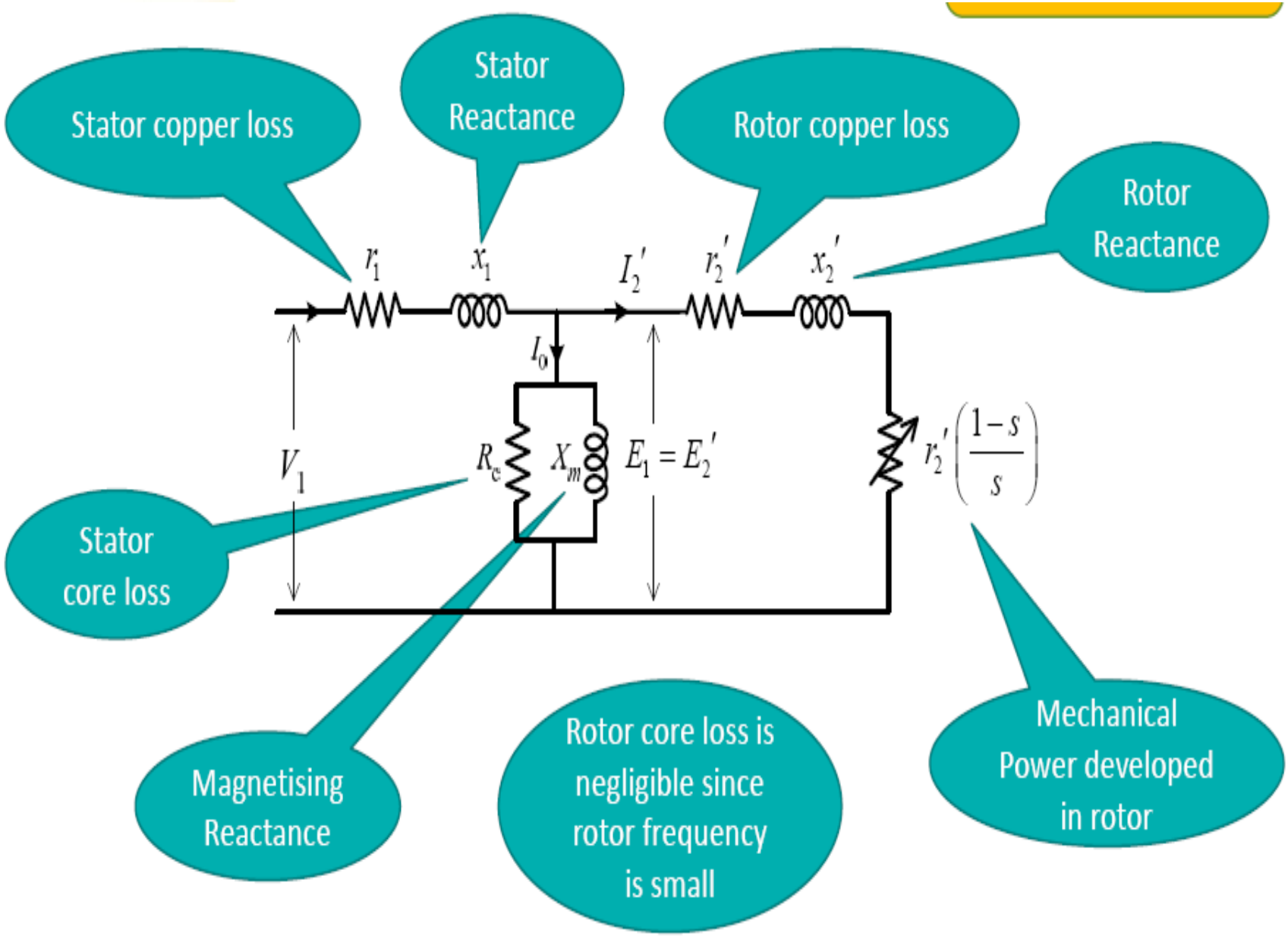
Induction Motors Equivalent circuit

- An induction motor can be described as rotating transformer, it is input is three phase voltage and current, the output of IM is shorted out so no electrical output exist, instead the output is mechanical.
- The Per phase equivalent circuit of an induction motor:



Rotor equivalent circuit

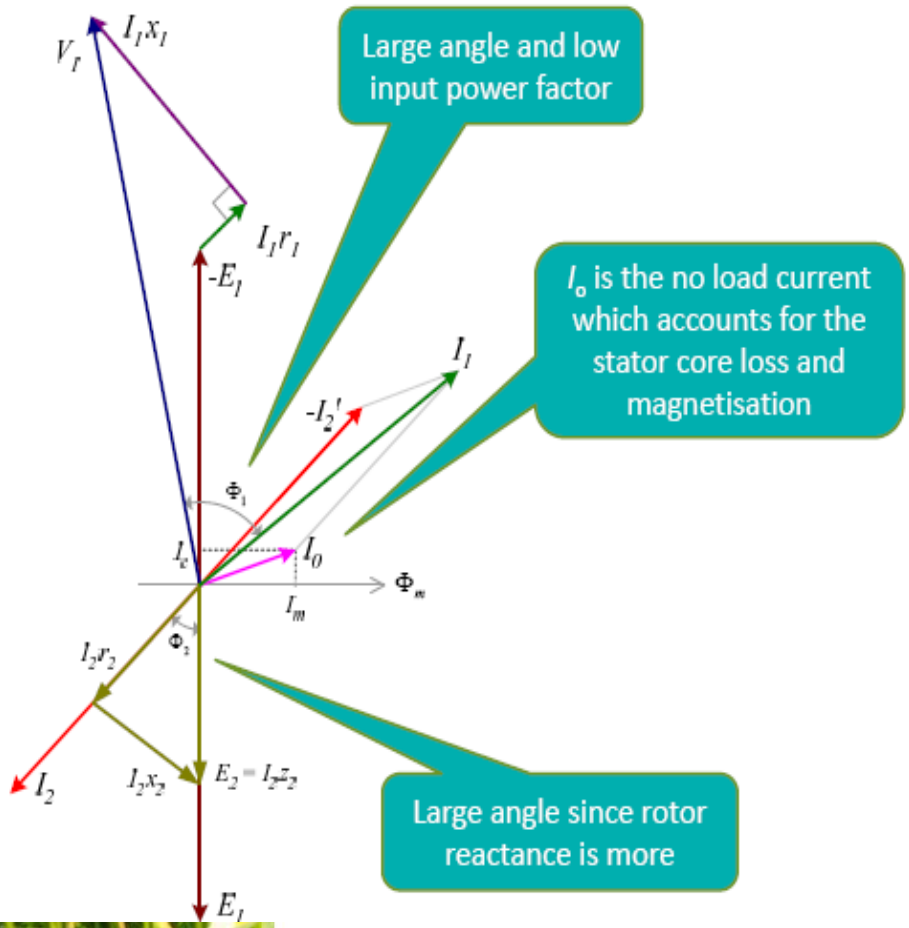




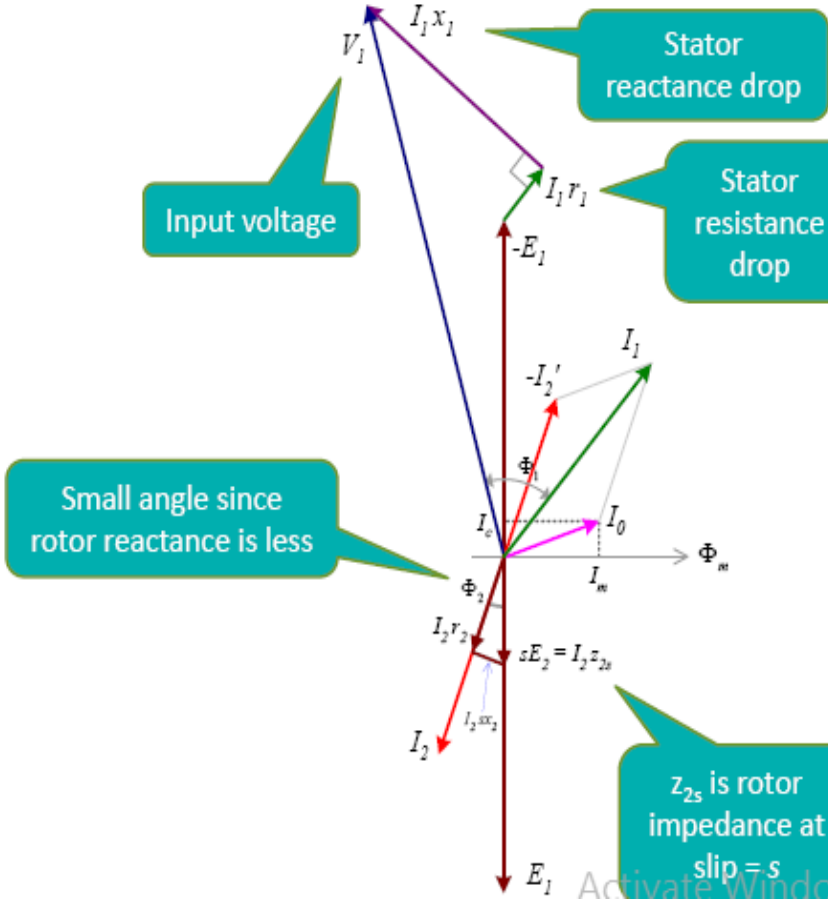
Activate Windows

Phasor Diagram

At Standstill



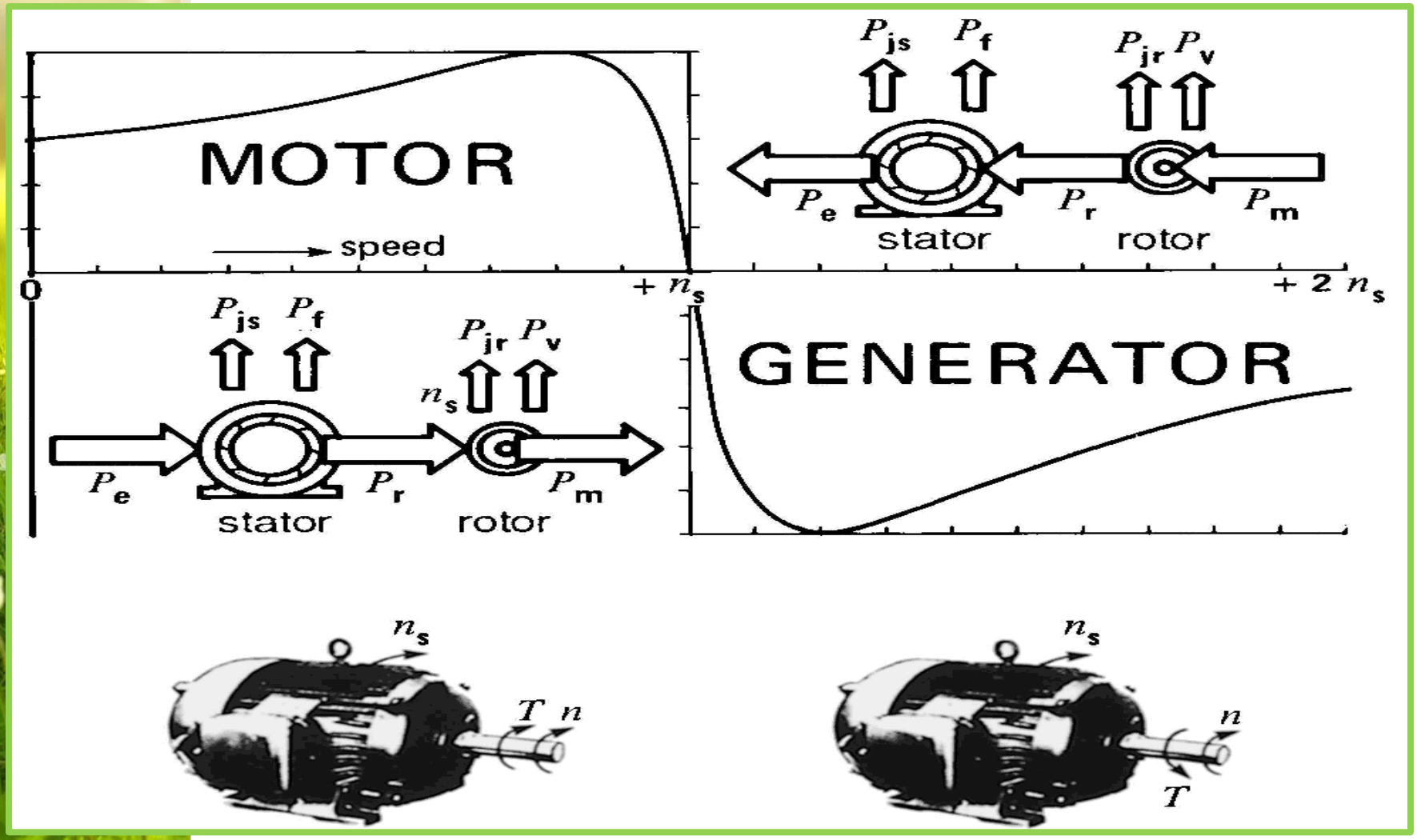
Operating at slip, *s*



Activate Windo



Induction Motor/Generator Mode





Thank you